

AI-ML Assignment – 4

Total Marks: 10

Topic

Breast Cancer Classification using K-Nearest Neighbors (KNN)



Submission Deadline

24 July 2026, 11:59 PM IST



Assignment PDF

Download the assignment from the shared Google Drive folder:

Google Drive Folder

https://drive.google.com/drive/folders/14_OGcBqYcEHc0F8Zfkezeqgf7LFcoDgF?usp=drive_link



Dataset

Breast Cancer Wisconsin Diagnostic Dataset

Kaggle Link:

<https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>

Problem Statement

A healthcare organization wants to develop a machine learning model to predict whether a breast tumor is **Malignant (M)** or **Benign (B)** based on diagnostic measurements.

Develop a **K-Nearest Neighbors (KNN)** classification model to classify tumors accurately.

Assignment Tasks (Total: 10 Marks)

Task 1: Data Understanding (2 Marks)

1. Load the dataset using Pandas.
 2. Display the first five records.
 3. Identify:
 - Numerical features
 - Target variable
 4. Display the dataset information and summary statistics.
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Task 2: Data Preprocessing (2 Marks)

Perform the following:

- Check for missing values.
- Remove unnecessary columns (if any).

- Encode the target variable if required.
 - Normalize or standardize the feature values.
 - Split the dataset into **80% training** and **20% testing**.
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Task 3: Model Development (3 Marks)

1. Train a **K-Nearest Neighbors (KNN)** classifier.
 2. Use **K = 5** as the initial value.
 3. Predict the class labels for the test dataset.
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Task 4: Model Evaluation (2 Marks)

Evaluate the model using:

- Accuracy Score
- Precision
- Recall
- F1-Score

Also generate:

- Confusion Matrix

Write **2–3 observations** based on the model performance.

Task 5: Conclusion (1 Mark)

Write a **100–150 word** conclusion covering:

- Key findings
 - Importance of feature scaling in KNN
 - One limitation of the KNN algorithm
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Submission Instructions

Create a **public GitHub repository** containing:

- `Assignment-4.ipynb` or `Assignment-4.py`
- `README.md` including:
 - Objective
 - Dataset Link
 - Libraries Used
 - Methodology
 - Results
 - Conclusion

Do not upload the dataset to GitHub unless its license explicitly allows redistribution. Instead, provide the Kaggle dataset link in your README.md.

Google Form Submission

Submit your assignment using the Google Form:

<https://forms.gle/fFL2CFooc5Vb2MXq8>

Fill in:

- Name

- Registration Number
 - Application Number
 - Batch Number
 - **Assignment Number (Assignment -4)**
 - Public GitHub Repository Link
 - Email Address
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Important Instructions

- **Deadline: 24 July 2026, 11:59 PM IST**
- Only the **first valid submission** will be accepted.
- Late submissions will be automatically rejected.
- Duplicate submissions will be automatically rejected.
- Future assignment submissions are not permitted.
- Ensure your GitHub repository remains **public** until evaluation is completed.
- You will receive an email confirming whether your submission has been **accepted** or **rejected**.